📊 Sales Performance Analysis

Data Analytics Internship Project

👷 Submitted By

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🏢 Internship Domain

Data Analytics

🎯 Project Objective

This project focuses on analyzing sales data to identify business trends, regional performance, profitable product categories, and seasonal sales patterns. The goal is to generate actionable business insights using data analytics and visualization techniques.

🛠️ Tools & Technologies Used

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Google Colab

📁 Dataset

Superstore Sales Dataset from Kaggle

📌 Key Analysis Performed

- Data Cleaning & Preprocessing
- KPI Analysis
- Monthly Sales Trend Analysis
- Regional Sales Analysis
- Category-wise Performance
- Top & Worst Performing Products
- Profitability Analysis
- Seasonal Sales Analysis
- Business Recommendations

📈 Expected Outcome

The analysis helps understand customer purchasing behavior, improve business decision-making, optimize inventory planning, and increase profitability through data-driven insights.

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

sns.set_style("whitegrid")
```

```
df = pd.read_csv('/content/superstore_final_dataset (1).csv', encoding='latin1')
```

```
df.head()
```

	Row_ID	Order_ID	Order_Date	Ship_Date	Ship_Mode	Customer_ID	Customer_Name	Segment
0	1	CA-2017-152156	8/11/2017	11/11/2017	Second Class	CG-12520	Claire Gute	Consumer
1	2	CA-2017-152156	8/11/2017	11/11/2017	Second Class	CG-12520	Claire Gute	Consumer
2	3	CA-2017-138688	12/6/2017	16/06/2017	Second Class	DV-13045	Darrin Van Huff	Corporate
3	4	US-2016-108966	11/10/2016	18/10/2016	Standard Class	SO-20335	Sean O Donnel	Consumer
4	5	US-2016-108966	11/10/2016	18/10/2016	Standard Class	SO-20335	Sean O Donnel	Consumer

Next steps: Generate code with df | New interactive sheet

```
df.shape
```

```
(9800, 18)
```

```
df.columns
```

```
Index(['Row_ID', 'Order_ID', 'Order_Date', 'Ship_Date', 'Ship_Mode',
       'Customer_ID', 'Customer_Name', 'Segment', 'Country', 'City', 'State',
       'Postal_Code', 'Region', 'Product_ID', 'Category', 'Sub_Category',
       'Product_Name', 'Sales'],
      dtype='object')
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 9800 entries, 0 to 9799
Data columns (total 18 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
```

```
 0   Row_ID         9800 non-null   int64
 1   Order_ID       9800 non-null   object
 2   Order_Date     9800 non-null   object
 3   Ship_Date      9800 non-null   object
 4   Ship_Mode      9800 non-null   object
 5   Customer_ID    9800 non-null   object
 6   Customer_Name  9800 non-null   object
 7   Segment        9800 non-null   object
 8   Country        9800 non-null   object
 9   City           9800 non-null   object
 10  State          9800 non-null   object
 11  Postal_Code    9789 non-null   float64
 12  Region         9800 non-null   object
 13  Product_ID     9800 non-null   object
 14  Category       9800 non-null   object
 15  Sub_Category   9800 non-null   object
 16  Product_Name   9800 non-null   object
 17  Sales          9800 non-null   float64
dtypes: float64(2), int64(1), object(15)
memory usage: 1.3+ MB
```

```
df.describe()
```

	Row_ID	Postal_Code	Sales
count	9800.000000	9789.000000	9800.000000
mean	4900.500000	55273.322403	230.769059
std	2829.160653	32041.223413	626.651875
min	1.000000	1040.000000	0.444000
25%	2450.750000	23223.000000	17.248000
50%	4900.500000	58103.000000	54.490000
75%	7350.250000	90008.000000	210.605000
max	9800.000000	99301.000000	22638.480000

Data Cleaning

```
df.isnull().sum()
```

	0
Row_ID	0
Order_ID	0
Order_Date	0
Ship_Date	0
Ship_Mode	0
Customer_ID	0
Customer_Name	0
Segment	0
Country	0
City	0
State	0
Postal_Code	11
Region	0
Product_ID	0
Category	0
Sub_Category	0
Product_Name	0
Sales	0

dtype: int64

```
df.duplicated().sum()
```

```
np.int64(0)
```

```
df['Order_Date'] = pd.to_datetime(df['Order_Date'], dayfirst=True)
```

```
df['Year'] = df['Order_Date'].dt.year
df['Month'] = df['Order_Date'].dt.month
df['Month_Name'] = df['Order_Date'].dt.month_name()
```

KPI Analysis

KPI 1 — Total Sales

```
total_sales = df['Sales'].sum()

print("Total Sales:", total_sales)
```

KPI 2 — Total Orders

```
total_orders = df['Order_ID'].nunique()

print("Total Orders:", total_orders)
```

KPI 3 — Total Customers

```
total_customers = df['Customer_ID'].nunique()

print("Total Customers:", total_customers)
```

KPI 4 — Average Order Value

Formula:

$$\text{Average Order Value} = \frac{\text{Total Sales}}{\text{Number of Orders}}$$

```
aov = df['Sales'].sum() / df['Order_ID'].nunique()

print("Average Order Value:", round(aov,2))
```

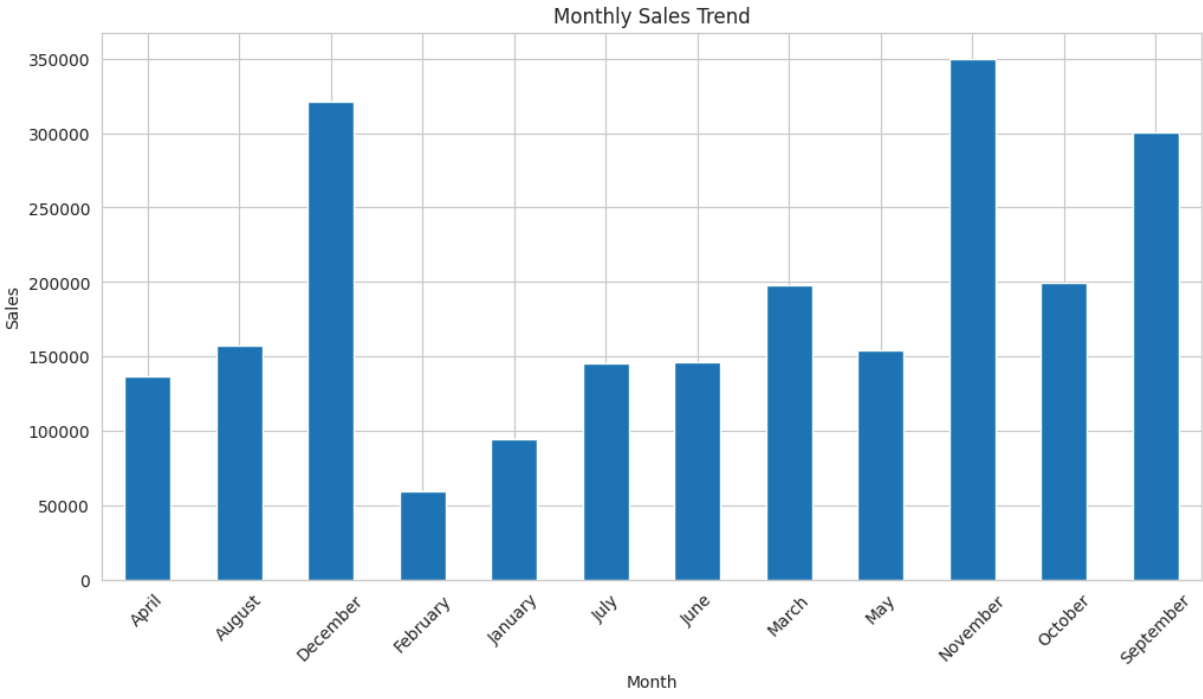
Note: The dataset did not include profit-related information. Therefore, the analysis primarily focused on sales trends, order patterns, customer distribution, and regional performance.

Insight

The KPI analysis provides a quick overview of the company's financial performance. Total sales indicate overall revenue generation, while profit margin helps measure operational efficiency and profitability.

Sales Trend Analysis

```
monthly_sales = df.groupby('Month_Name')['Sales'].sum()

plt.figure(figsize=(12,6))

monthly_sales.plot(kind='bar')

plt.title('Monthly Sales Trend')
plt.xlabel('Month')
plt.ylabel('Sales')

plt.xticks(rotation=45)

plt.show()
```

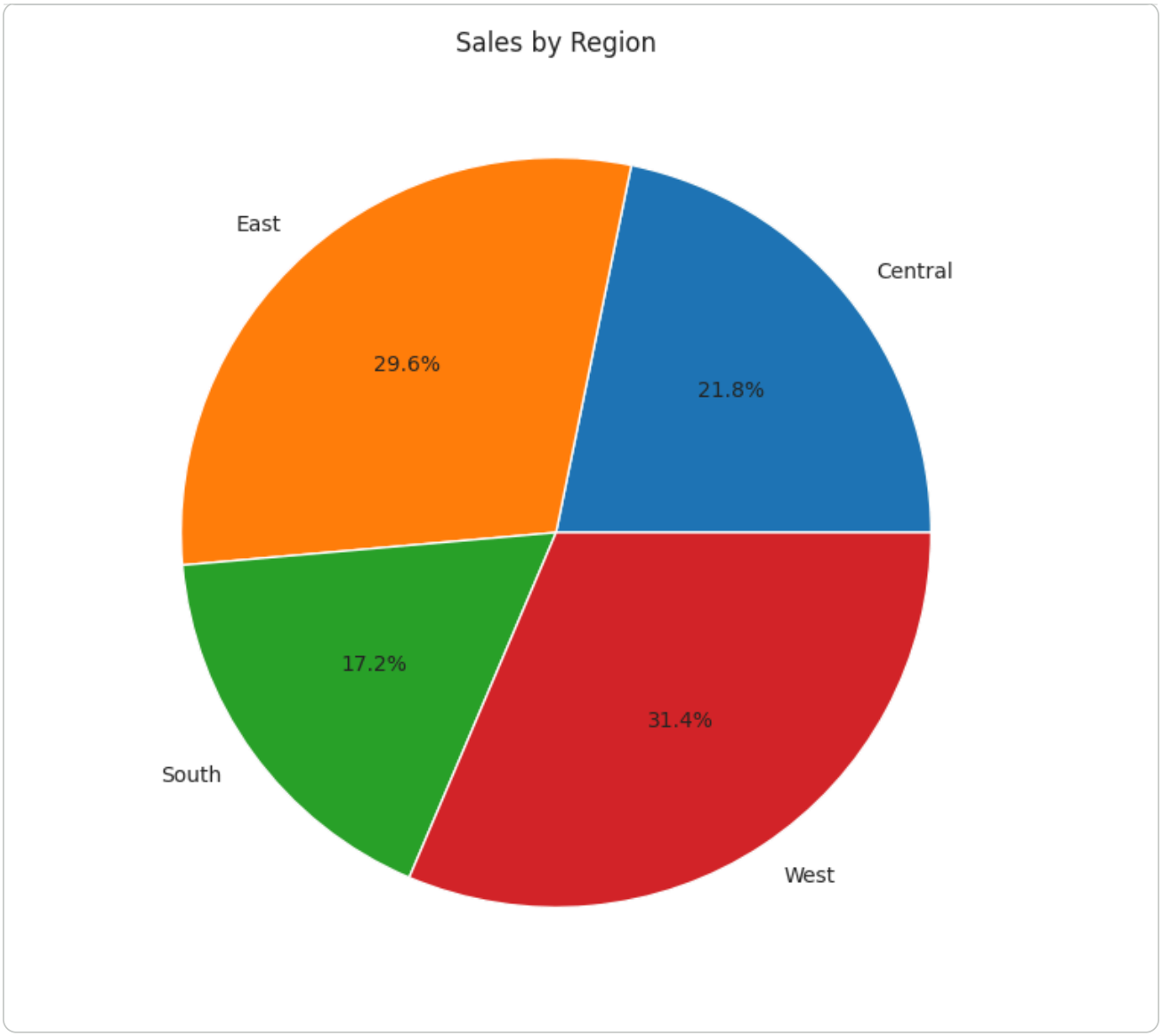


Insight

The monthly sales trend shows fluctuations in revenue across different months. Certain months generated significantly higher sales, indicating possible seasonal demand and peak purchasing periods.

Region-wise Sales Analysis

```
region_sales = df.groupby('Region')['Sales'].sum()

plt.figure(figsize=(8,8))

region_sales.plot(kind='pie', autopct='%1.1f%%')

plt.title('Sales by Region')

plt.ylabel('')

plt.show()
```

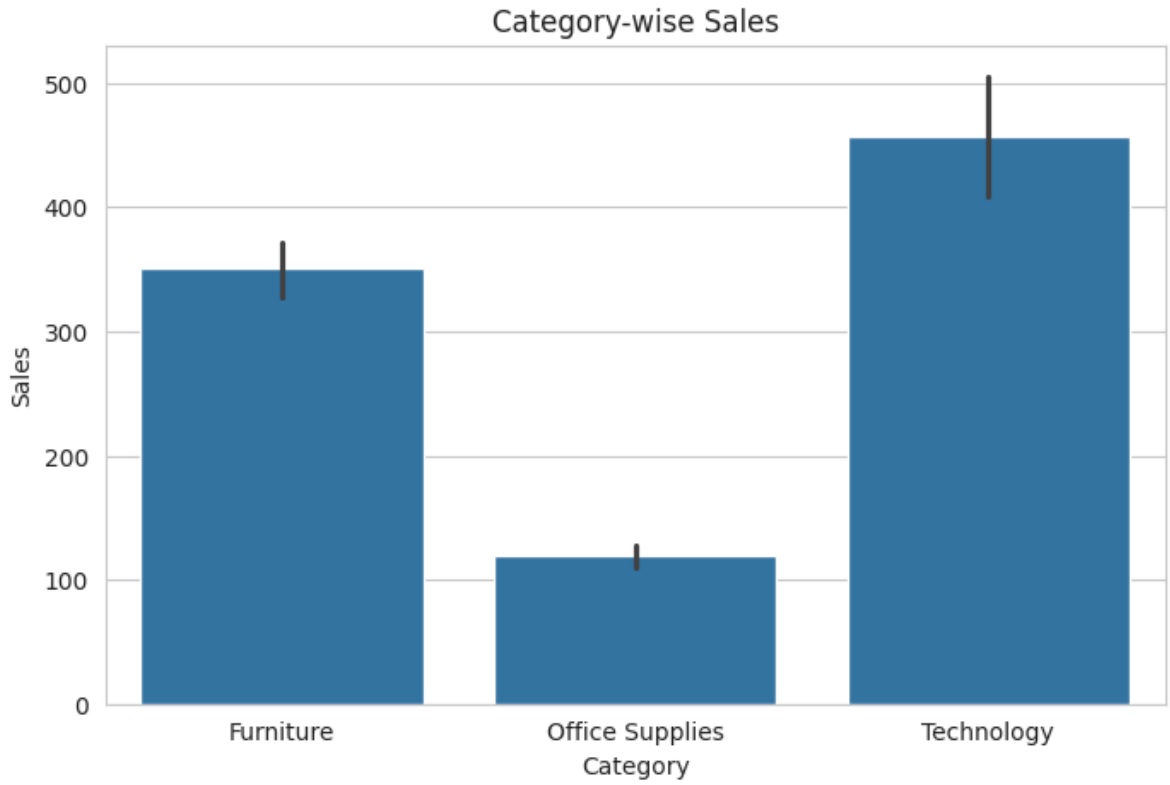


Insight

Regional analysis reveals that some regions contribute more revenue than others. High-performing regions may have stronger customer demand, while low-performing regions may require targeted marketing and promotional strategies.

Category-wise Sales

```
plt.figure(figsize=(8,5))

sns.barplot(x='Category', y='Sales', data=df)

plt.title('Category-wise Sales')

plt.show()
```

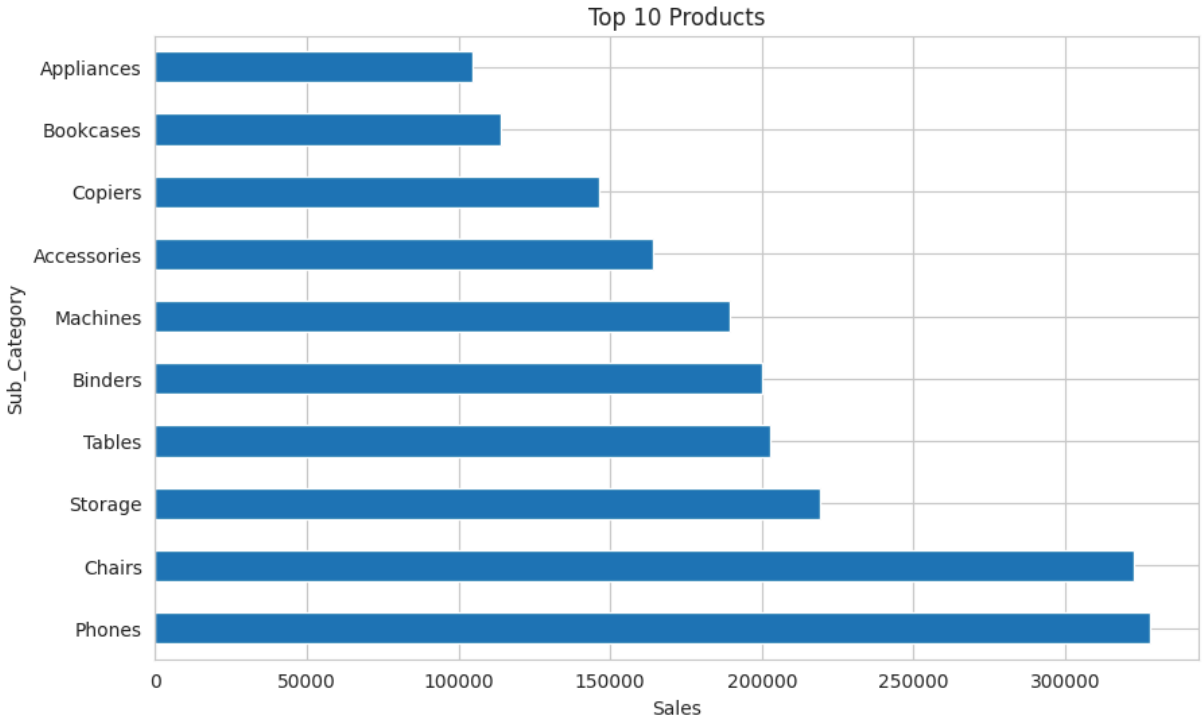


Insight

The category analysis shows which product categories dominate overall sales. Categories with higher contribution indicate strong customer preference and market demand.

Top 10 Selling Products

```
top_products = df.groupby('Sub_Category')['Sales'].sum().sort_values(ascending=False).head(1

plt.figure(figsize=(10,6))

top_products.plot(kind='barh')

plt.title('Top 10 Products')

plt.xlabel('Sales')

plt.show()
```

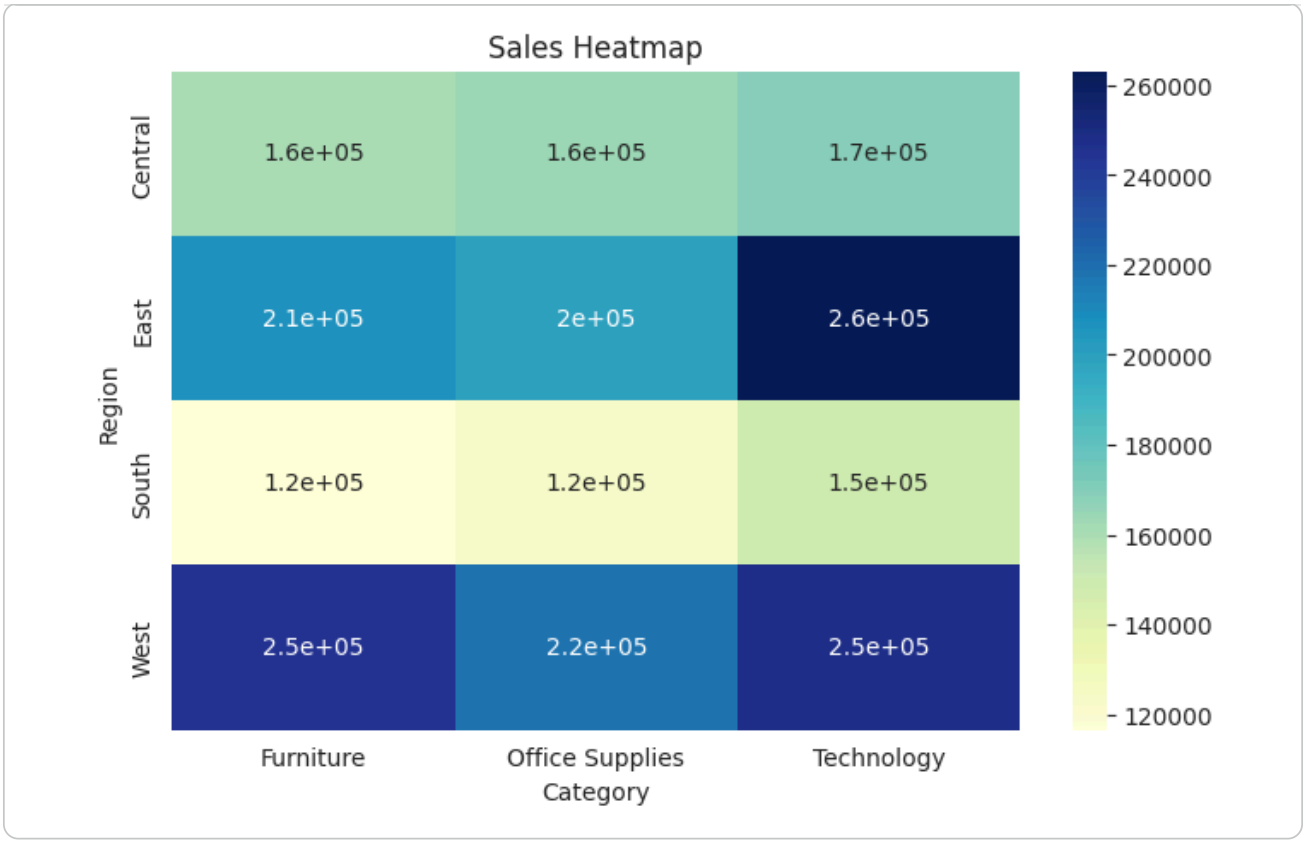


Business Insight

The analysis identifies the best-selling sub-categories driving revenue growth. Maintaining product availability and customer engagement for these products can improve long-term sales performance.

Heatmap Analysis

```
pivot = df.pivot_table(
    values='Sales',
    index='Region',
    columns='Category',
    aggfunc='sum'
)

plt.figure(figsize=(8,5))

sns.heatmap(pivot, annot=True, cmap='YlGnBu')

plt.title('Sales Heatmap')

plt.show()
```

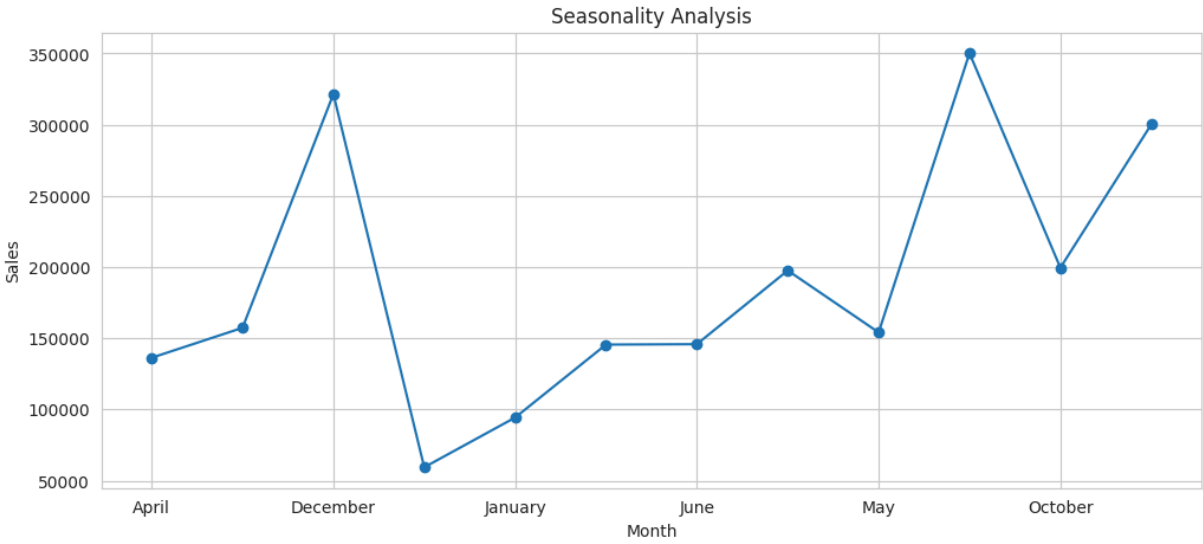


Insight

The correlation analysis helps identify relationships between numerical variables such as sales, profit, quantity, and discounts. Strong correlations can support better business decision-making and forecasting.

Seasonality Analysis

```
seasonality = df.groupby('Month_Name')['Sales'].sum()

plt.figure(figsize=(12,5))

seasonality.plot(marker='o')

plt.title('Seasonality Analysis')

plt.xlabel('Month')

plt.ylabel('Sales')

plt.grid(True)

plt.show()
```

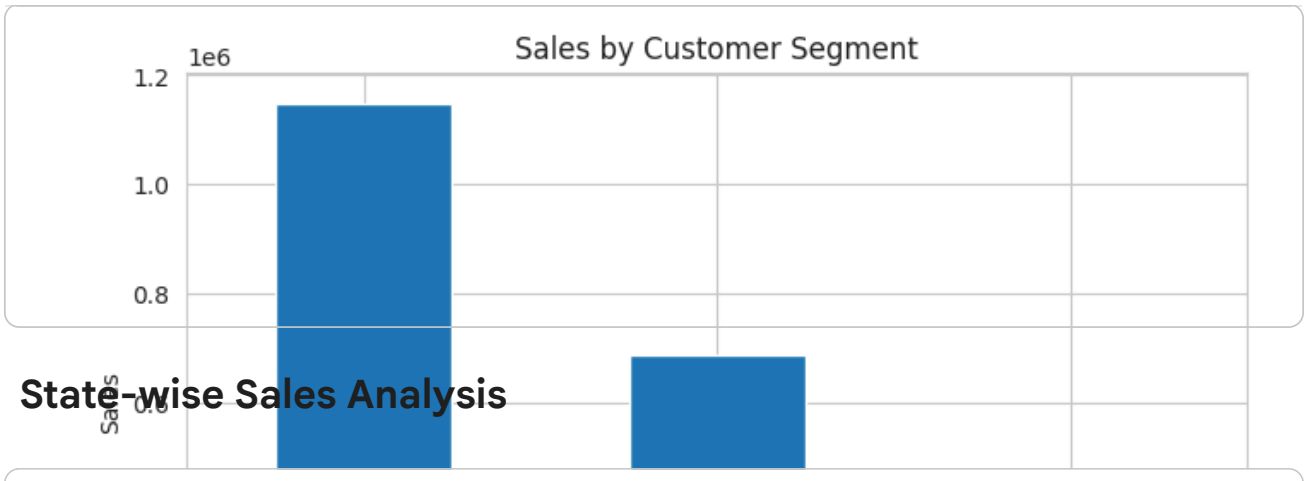


Business Insight

Seasonal analysis indicates recurring sales patterns throughout the year. Understanding these trends helps businesses optimize inventory planning, staffing, and marketing strategies.

Customer Segment Analysis

```
segment_sales = df.groupby('Segment')['Sales'].sum()

plt.figure(figsize=(8,5))

segment_sales.plot(kind='bar')

plt.title('Sales by Customer Segment')

plt.xlabel('Segment')

plt.ylabel('Sales')

plt.show()
```



State-wise Sales Analysis

```
top_states = df.groupby('State')['Sales'].sum().sort_values(ascending=False).head(10)

plt.figure(figsize=(10,6))

top_states.plot(kind='bar')

plt.title('Top 10 States by Sales')

plt.xlabel('State')

plt.ylabel('Sales')

plt.xticks(rotation=45)

plt.show()
```